

Create my YouTube playlist: Foundations of Mathematics

Build an ordered, rigorous learning path from formal logic and proof to set theory, computability, Goedel incompleteness, type theory, and category theory.

1. Primary task

Use Cloud Browser on my YouTube account. Create one playlist named: Foundations of Mathematics - Logic to Sets to Goedel

- 1 Search YouTube for each item in the ordered curriculum below.
- 2 Prefer the exact channel and title listed. Verify the uploader before adding.
- 3 Add the videos in the exact pedagogical order below.
- 4 Do not add duplicates, Shorts, reaction videos, reuploads, or low-quality summaries.
- 5 For full university courses, add only the listed relevant lectures unless the curriculum explicitly says to add the full playlist.
- 6 If an exact video has disappeared, substitute a high-quality equivalent from the same institution/channel where possible.
- 7 After creation, review the playlist order once from top to bottom.

2. Learner profile and objective

The learner is a Danish HTX student with strong interest in mathematics, programming, computer science, and rigorous explanations. The material may be university level, but it should build progressively and not assume advanced real analysis.

The target is not merely popular mathematics. By the end, the learner should be able to explain the chain:

logic -> axioms -> sets -> number systems -> formal theories -> computation -> undecidability -> incompleteness

and understand what statements such as "ZFC proves X", "a theory is consistent", "a problem is undecidable", and "Goedel incompleteness" actually mean.

3. Fourteen-week study structure

Week	Core topic	Learning target
1	Propositional logic	Truth tables, implication, equivalence, negation.
2	Quantifiers and proof	Direct proof, contradiction, contraposition, cases.
3	Induction	Ordinary and strong induction; recursive definitions.
4	Sets and functions	Power sets, products, injections, surjections, bijections.
5	Constructing natural numbers	Understand numbers as mathematical objects built from sets.
6	Integers, rationals, reals	See how familiar number systems can be constructed.
7	Infinity	Countability, uncountability, Cantor diagonalization.
8	Axiomatic set theory	Russell's paradox and the purpose of ZFC axioms.
9	Formal systems	Syntax vs semantics; axioms, inference rules, theorem, truth.

Week	Core topic	Learning target
10	Computability	Turing machines and Church-Turing thesis.
11	Undecidability	Halting problem, diagonalization, reductions.
12	Goedel incompleteness	Goedel numbering, self-reference, first and second theorem.
13	Type theory	Propositions-as-types and proof/program connections.
14	Category theory and synthesis	Categories, morphisms, functors; connect the entire map.

Ordered YouTube curriculum

Playlist contents

Search using both the title and channel. Keep this order unless a video's internal course ordering requires adjacent parts.

Phase A - Logic, proof, sets, and numbers

1. The Bright Side of Mathematics - Start Learning Mathematics

Search: The Bright Side of Mathematics Start Learning Mathematics

Add the introductory video and the relevant sequence covering logic, sets, natural numbers, integers, rationals, reals, and complex numbers. If there is an official playlist, prefer it.

2. MIT OpenCourseWare - 6.042J Mathematics for Computer Science

Search: MIT 6.042J Mathematics for Computer Science proofs logic induction sets

Add the relevant lectures on introduction to proofs, contradiction, proof by cases, well ordering, logic/propositions, quantifiers, sets, induction, and infinite sets. Do not add the entire course unless needed to preserve a coherent lecture sequence.

3. TrevTutor - Direct Proof / discrete mathematics proof videos

Search: TrevTutor Direct Proof discrete mathematics

Add a concise direct-proof lesson as reinforcement after the MIT proof material. If useful, add one video on contraposition and one on contradiction from the same series.

4. Oxford Mathematics - Set Theory: What is Set Theory and what is it for?

Search: Oxford Mathematics Set Theory What is Set Theory and what is it for

Add the official Oxford Mathematics upload.

5. Oxford Mathematics - Set Theory: Russell's Paradox

Search: Oxford Mathematics Set Theory Russell's Paradox

Add directly after the introductory Oxford set theory lecture.

6. ZFC overview - concise axioms introduction

Search: 10 Axioms in 10 Minutes ZFC Set Theory

Add a reputable concise overview of the ZFC axioms. Prefer a mathematically careful explanation over a purely entertainment-oriented one.

Phase B - Computability and undecidability

7. MIT OpenCourseWare - 18.404J Theory of Computation, Michael Sipser

Search: MIT 18.404J Theory of Computation Michael Sipser

Add the official MIT lectures corresponding to these topics, in this order:

- 1 Turing Machines
- 2 Church-Turing Thesis
- 3 Decision Problems
- 4 Undecidability
- 5 Reducibility
- 6 Recursion Theorem and Logic

Phase C - Goedel and limits of formal systems

8. Numberphile - Goedel's Incompleteness Theorem

Search: Numberphile Godel Incompleteness Theorem

Use as intuition/motivation, not as the final rigorous treatment.

9. Goedel's First Incompleteness Theorem - proof / diagonalization

Search: Godel 1st Incompleteness Theorem Proof by Diagonalization

Add a detailed lecture that explains encoding, self-reference, and diagonalization. Prefer a formal mathematical treatment.

10. Joel David Hamkins - The Goedel incompleteness phenomenon

Search: Joel David Hamkins Godel incompleteness phenomenon

Add the official or institution-hosted lecture. This is the deeper capstone for incompleteness.

Alternative foundations and unification

Phase D - Type theory

11. Peter Dybjer - Intuitionistic Type Theory, Lecture I

Search: Peter Dybjer Intuitionistic Type Theory Lecture I

Add the first introductory lecture. If an official short series follows naturally, add the next 1-2 lectures only if they remain introductory.

12. Curry-Howard correspondence

Search: Curry Howard correspondence propositions as types proofs as programs lecture

Add one rigorous but accessible lecture explaining the propositions-as-types / proofs-as-programs correspondence. Prefer a university or recognized mathematics/computer science educator.

13. Homotopy Type Theory - optional extension

Search: Homotopy Type Theory introduction foundations of mathematics

Add one introductory HoTT video after ordinary type theory. Treat it as optional/advanced; do not place it before basic type theory.

Phase E - Category theory

14. Bartosz Milewski - Category Theory for Programmers

Search: Bartosz Milewski Category Theory for Programmers playlist

Add approximately the first 4-6 lectures, starting with the basics of categories, morphisms, composition, and functors. Do not add the entire long course unless there is no practical way to select the introductory block.

4. Quality rules for substitutions

- Priority order for sources: MIT/OpenCourseWare, Oxford or other universities, recognized researchers/lecturers, then strong specialist educators.
- Prefer lectures with equations, definitions, proofs, and worked examples.
- Avoid videos whose main purpose is "mind-blowing math facts" without formal content.
- English content is preferred; Danish is acceptable only if equal or better in rigor.
- For ambiguous Goedel results, prefer content that clearly distinguishes truth, provability, consistency, completeness, and effective axiomatizability.

5. Final playlist check

Before finishing, verify that the playlist broadly follows this progression:

Logic -> proof -> induction -> sets/functions -> construction of numbers -> infinity -> ZFC -> formal systems -> Turing machines -> undecidability -> Goedel -> type theory -> category theory

Make sure no advanced Goedel, type theory, or category theory lecture appears before its prerequisites.

6. Suggested final project for the learner

After completing the playlist, the learner should be able to produce a rigorous explanation of why $1 + 1 = 2$ starting from definitions and axioms:

logic -> set-theoretic definitions of natural numbers -> successor -> recursive definition of addition -> derivation of $1 + 1 = 2$

Then explain why the deeper foundational question is not merely whether a statement can be proved, but what "proof" means inside a formal system and what the inherent limits of such systems are.

7. Completion instruction for Work

When the playlist is complete, report: (1) the playlist name, (2) total number of videos, (3) any substitutions made because an exact requested video could not be found, and (4) whether the ordering was verified.

Prepared as a handoff document for ChatGPT Work / Cloud Browser. The Work agent should verify current YouTube availability and exact uploader identity at execution time.